physics modelling agent

progress report

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supervisor 2: Prof. Dr. Samir Lounis

12th July, 2026

outline

aim of the project

what the "physics modelling agent" should be

A configuration of "intelligent systems" which assists a human researcher in building a model to answer a research question.

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reports findings:
- human readable **model description**
- executable **numerical implementation**

definitions of LLM and agentic AI

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- “Transformer language models” [19] with at least billions of parameters (no clear cut-off) [19]
- mostly use tokenization: dividing text into pieces and applying transformation function

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what is an agentic AI

- an AI with enhanced capabilities including: strategic planning, (external) tools usage, collaboration with other agents [3]

architectural foundations

constraints

- only use open source software
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  - access via remote servers with valid API key [9]

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- use an existing framework for AI orchestration
  - save time on the software engineering part of the project
  - focus on definition of AI agents and configuration of their behaviour

Figure: framework chosen: crewai [6]

crewai agents

Agents[4]

accesses an agentic AI model to perform a task

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- allows to access any AI-models we have access to (SAIA)
- behaviour can be defined via parameters (role, goal, backstory, creativity)
- tools can be added to enhance capabilities (e.g. web search, code execution)

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Tasks[7]

describe a concrete action to execute

- defined by task description and expected output
- is assigned an agent (execution merges agent and task description)

crewai: orchestration

the Crew[5]

contains all agents with their tasks and handles the interactions between agents

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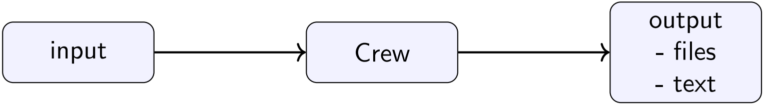

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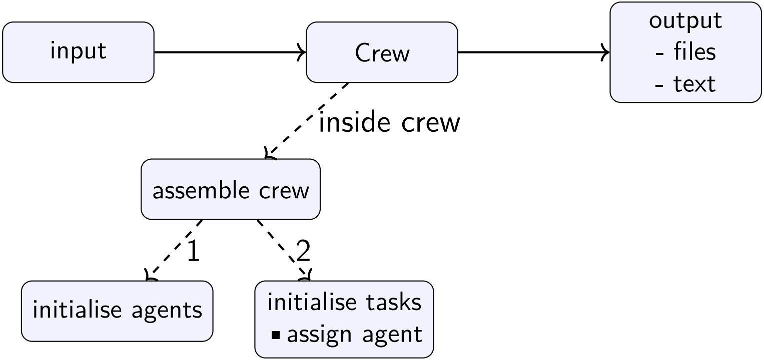

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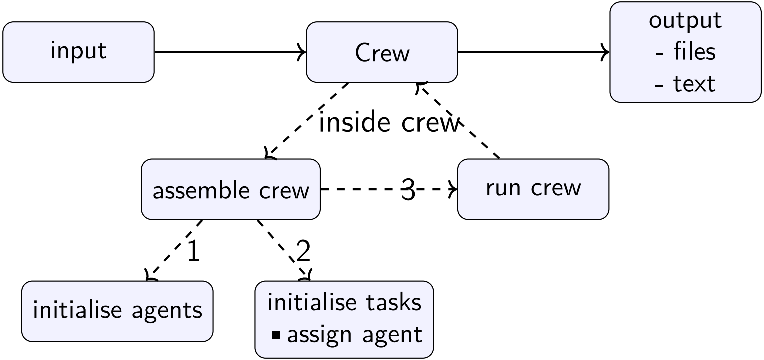

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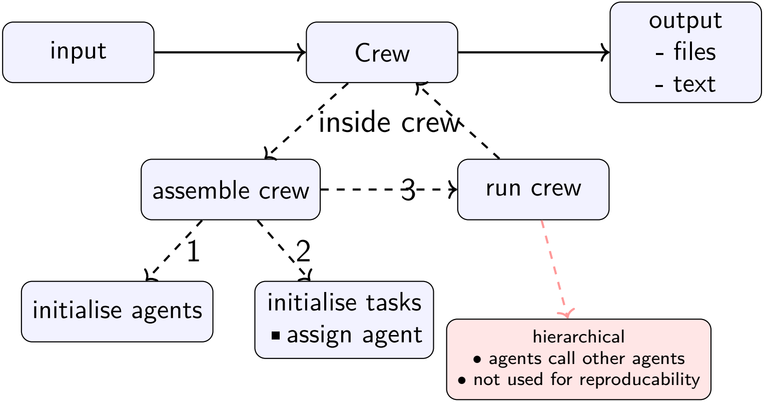

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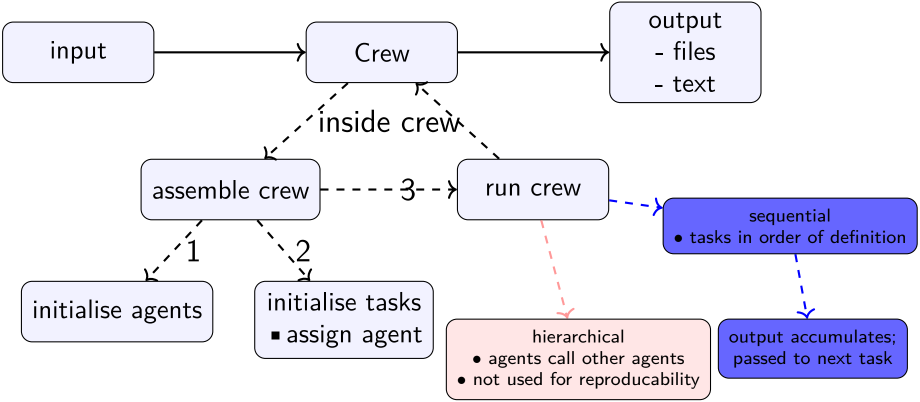

general design of the pipeline

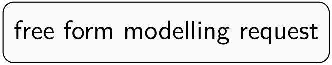

Figure: general vision for the steps the crew should perform

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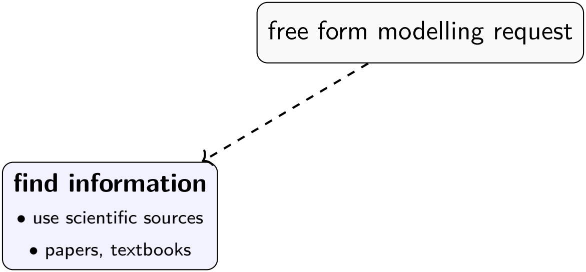

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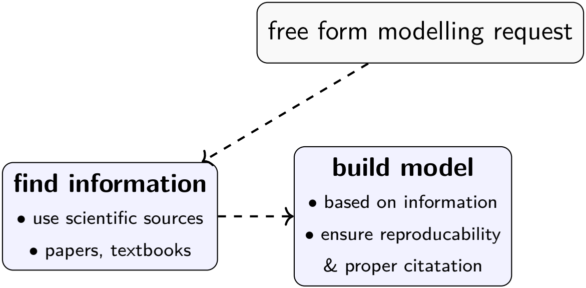

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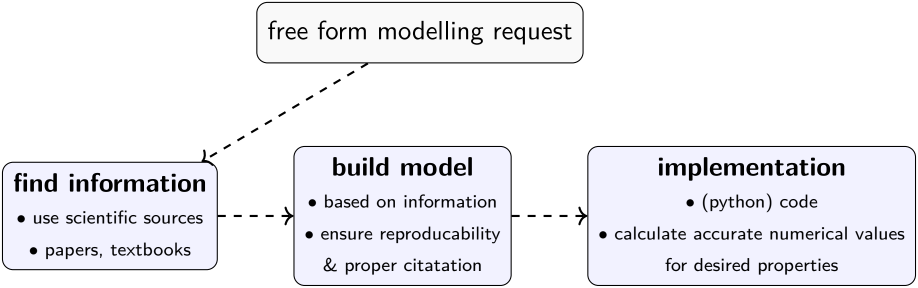

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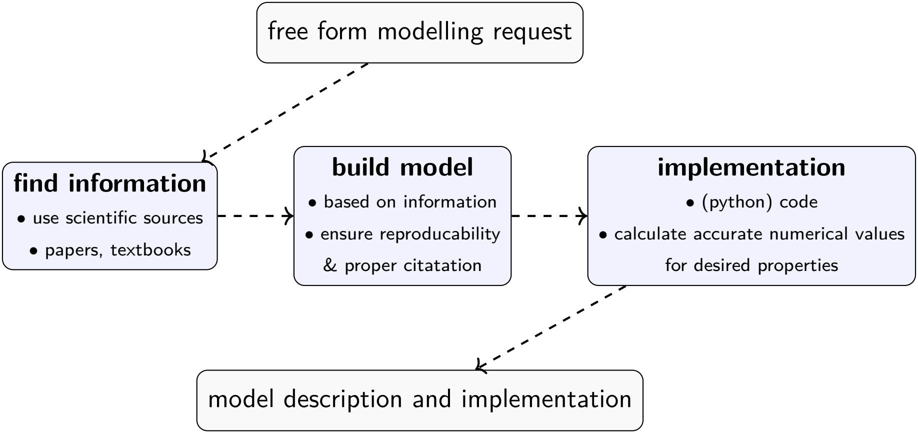

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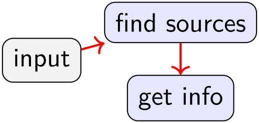
information finding



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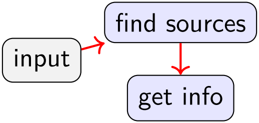

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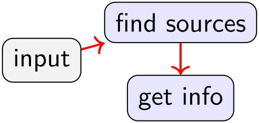

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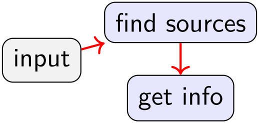

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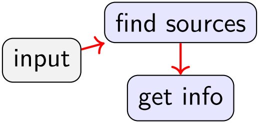

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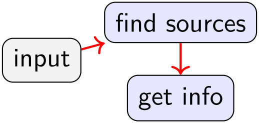

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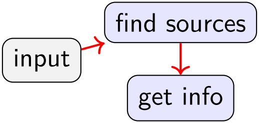

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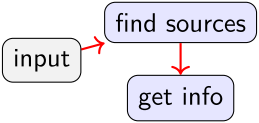

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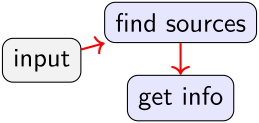

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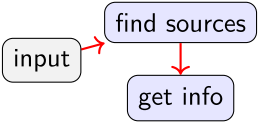
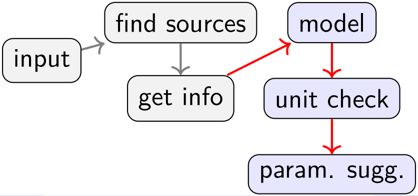
✓ implement custom tools performing exponential back-off

✓ separate the task performing the arXiv request from the crew

$\Longrightarrow$ no new requests when testing other tasks



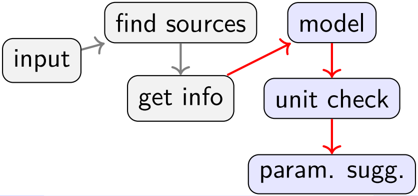
modelling



modelling

the unit problem

1. task outputs formula from the sources; theoretical formulas often use different definitions of parameters



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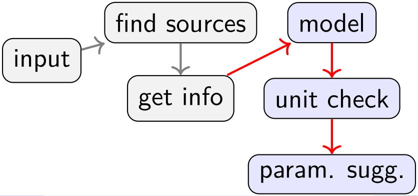

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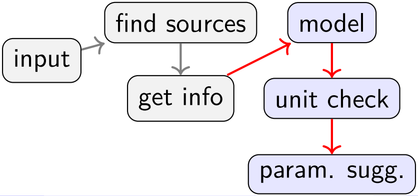

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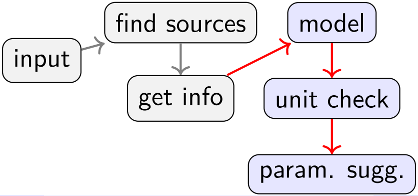

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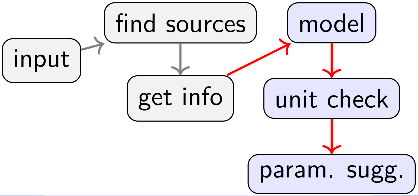
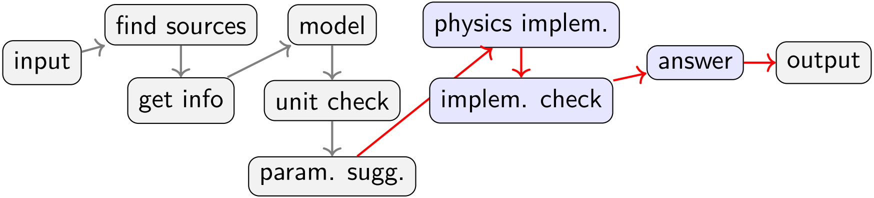
$\Longrightarrow$ agent for suggesting reasonable starting parameters



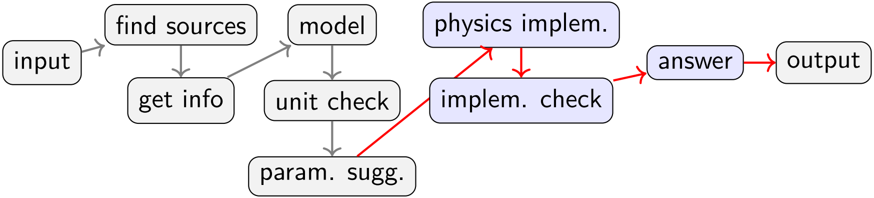
numerical implementation



numerical implementation

minor challenges

1. first simulation agent specialized on logic often puts out code with bugs



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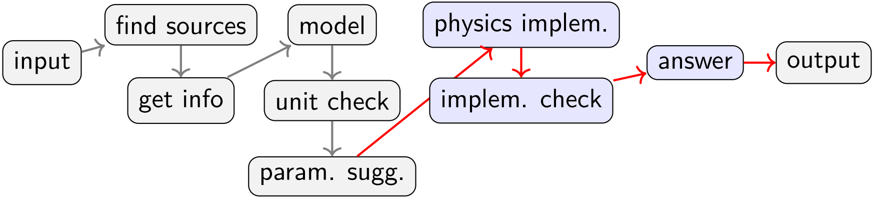

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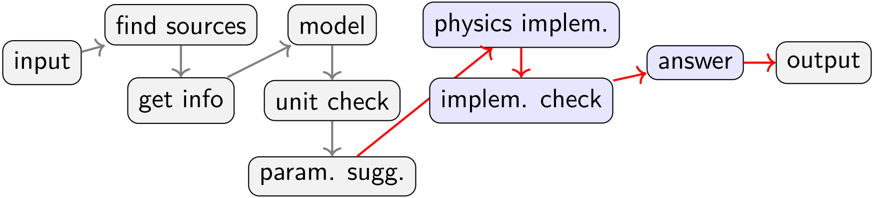
answer: provide a short result at runtime

- needed for automatic evaluation of the system (benchmarking)



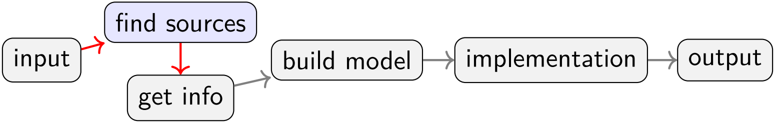
example: input/modelling request

input

aim : "Calculate the Edelstein effect for a Rashba fermion (at the Gamma point of the Brillouin zone).
Compute the magnitization magnitude and direction of different directions and magnitudes of the applied electric field.
Consider how the result depends on relevant parameters of the model (e.g. chirality, fermi velocity, spin-orbit coupling strength) and make explicit graphics."

topic : Edelstein-Effect

papers found



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agent configuration:

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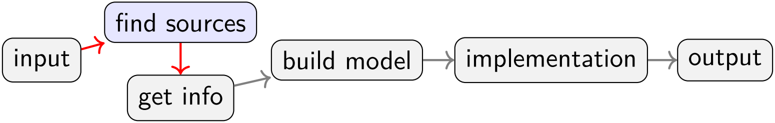
- tools: arXiv_Papers_With_Backoff



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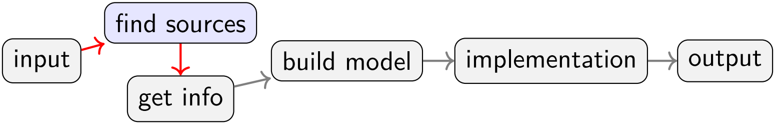
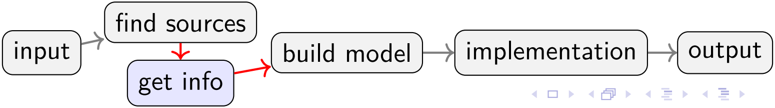
papers found (view full report)

"Edelstein Effect in Isotropic and Anisotropic Rashba Models"[12]
"Spin and orbital Edelstein effect in spin-orbit coupled noncentrosymmetric superconductors"[1]
"Spin and orbital Edelstein effect in a bilayer system with Rashba interaction"[14]



information extraction



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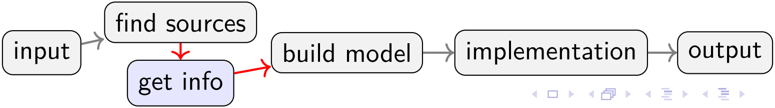

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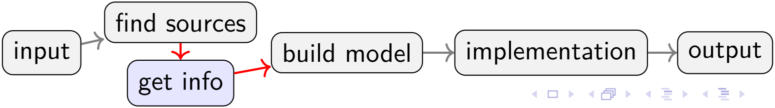
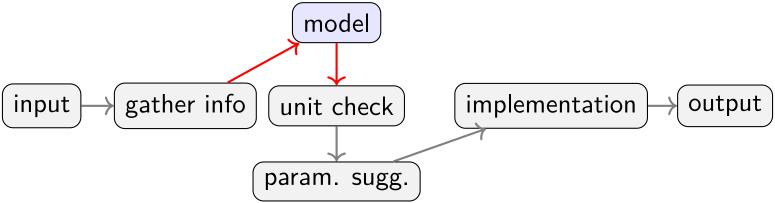
extracted information (view full report)

$\hat{H} = \frac{p^2}{2m} + \alpha \hat{z} \cdot (\vec{p} \times \vec{\sigma})$	Rashba-Hamiltonian
$M_y = \frac{\mu_b \lvert e \rvert \tau}{2\pi} m\alpha \left[\hat{z} \times \mathbf{E}\right]_y$	magnetization in HDR
$M_y = \frac{\mu_b \lvert e \rvert \tau}{2\pi} \sqrt{m^2\alpha^2 + 2mE_F} \left[\hat{z} \times \mathbf{E}\right]_y$	magnetization in LDR



build model


model
input
gather info
unit check
implementation
output
param. sugg.

build model

agent configuration:

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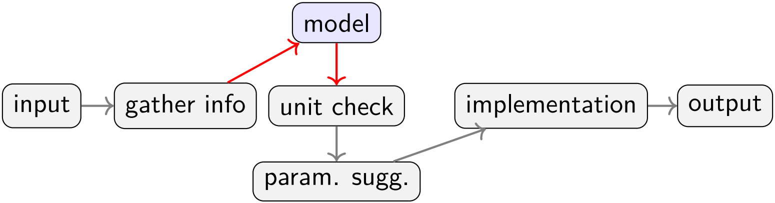
- tools: this LLM is unable to use tools



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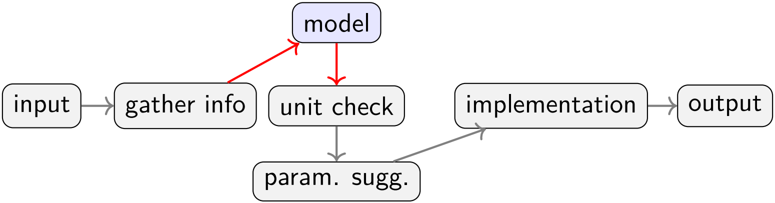
extracted information (view full report)

restates results from information extraction



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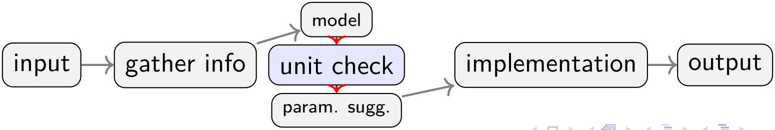
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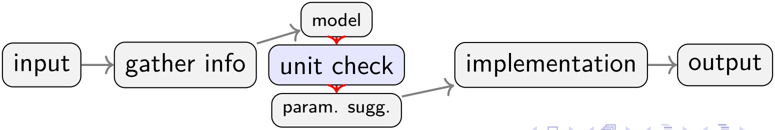

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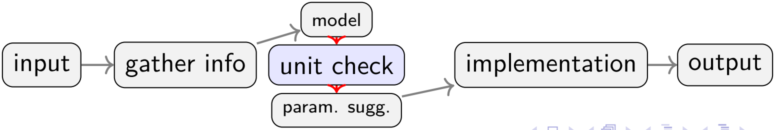
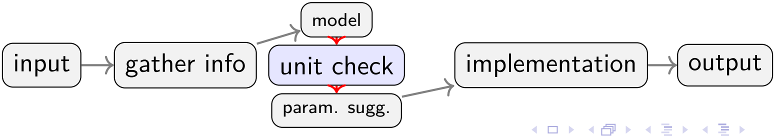
$$\frac{E}{BL^3} = \frac{E}{B} \cdot Q \cdot T \cdot M \cdot EL \cdot \frac{E}{QL}$$ dimensions

$$\Longleftrightarrow \quad \frac{E}{BL^3} = E^3 \cdot M \frac{T}{B}$$

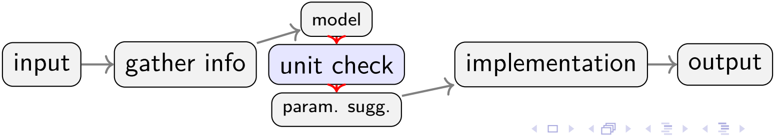


unit check


model
input
gather info
unit check
implementation
output
param. sugg.

unit check

- little creativity (temperature $= 0.2$)
- LLM model: qwen3.5-122b-a10b[16]; (allow tool usage)
- tools: dimensional_Analysis using SymPy [17]



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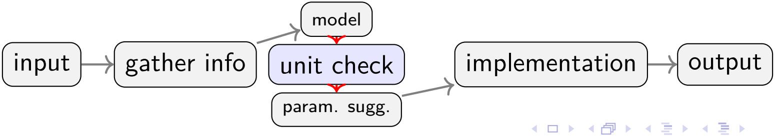
unit corrections (view full report)

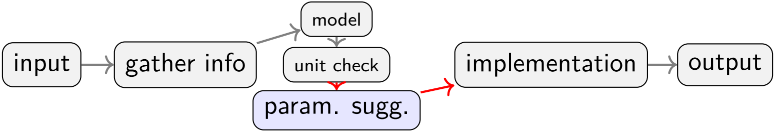
$$\hat{H} = \frac{p^2}{2m} + \frac{\alpha}{\hbar}\hat{z}\cdot(\vec{p}\times\vec{\sigma})$$ Rashba-Hamiltonian

$$M_y = \frac{\mu_b|e|\tau}{2\pi\hbar^2} m\alpha\left[\hat{z}\times\mathbf{E}\right]_y$$ magnetization in HDR

$$M_y = \frac{\mu_b|e|\tau}{2\pi\hbar^2}\sqrt{m^2\alpha^2 + 2mE_F}\left[\hat{z}\times\mathbf{E}\right]_y$$ magnetization in LDR

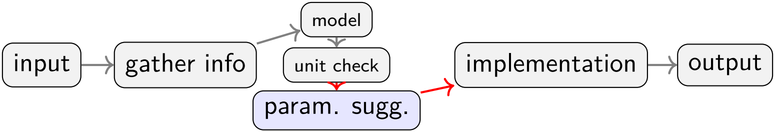


parameter suggestion



parameter suggestion

- no creativity (temperature = 0)
- LLM model: qwen3.5-122b-a10b[16]; (general purpose)
- tools: no tool used



parameter suggestion

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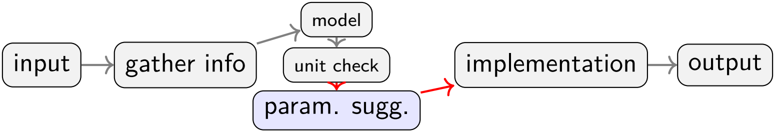
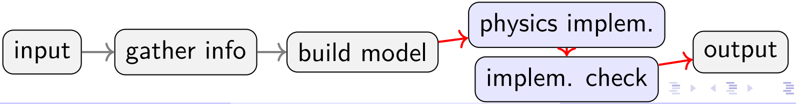
suggested starting parameters (view full report)

$m \approx 6,10 \times 10^{-32}\text{kg}$	effective mass
$\alpha = 5,0 \times 10^{4}\text{m/s}$	Rashba Coupling Strength
$\tau = 5,0 \times 10^{-13}\text{s}$	transport time
$E_x = 1,0 \times 10^{5}\text{V/m}$	applied electric field



numerical implementation I



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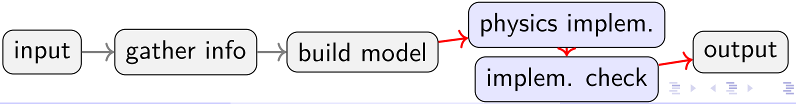

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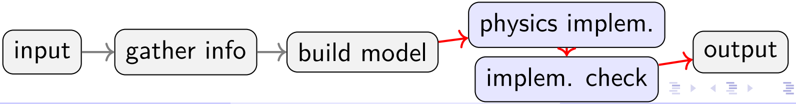
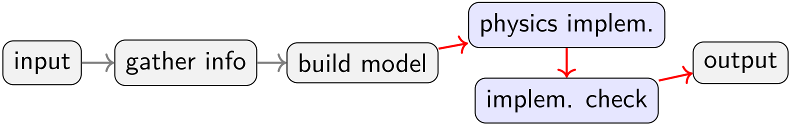
results for magnetisation (view full report)

Isotropic HDR Magnetization:	$M_y = 3,25 \times 10^{-9}$ A/m
Isotropic LDR Magnetization:	$M_y = 1,52 \times 10^{-8}$ A/m
Anisotropic Magnetization Components:	$M_x = 2,04 \times 10^{-8}$ A/m
	$M_y = 2,04 \times 10^{-8}$ A/m

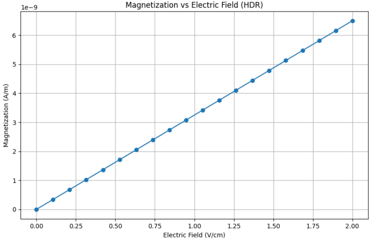
some minor corrections necessary



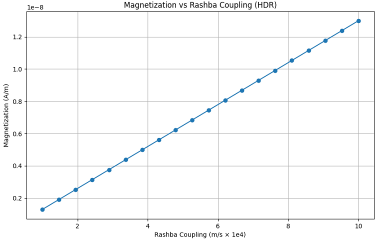
numerical implementation II



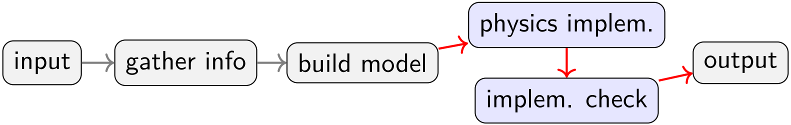
numerical implementation II



(a) magnetisation over electric field



(b) magnetisation over Rashba param. α



next steps

benchmark the “physics-modelling-agent”

a method to measure the systems performance

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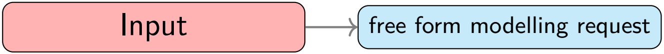
3. output variance: how consistent is the system over multiple runs?
4. execution time: how is it influenced by parameters?
5. sources: how does the number (and possibly type) of sources influence the systems results?

summary



summary

TASK
- description
- expected output
- agent

AGENT
- role, goal, backstory
- LLM model, temperature
- assigned tools

Input → free form modelling request

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↓

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suggest a model → based on given information

↓

check & correct units → dimensional analysis

↓

implement model → executable python code calculations

sources I

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